Technology Stack

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

The stack is entirely open source and chosen so that the whole system runs offline on a laptop without a GPU.

1. Stack Overview by Layer

Layer	Technology	Role
Language / runtime	Python 3.11.12 (CPython)	Primary language for all services
Computer vision	OpenCV 4.11.0	Frame capture, Haar cascade face detection, solvePnP, annotations
Landmarks	MediaPipe 0.10.35	478-point face mesh including iris landmarks
Tracking	DeepSort Realtime 1.3.2	Kalman tracking with a 128-d color + spatial appearance embedding
Recognition	DeepFace 0.0.102 (Facenet)	128-d embeddings, cosine distance matching
Audio	Librosa 0.10.2, SoundFile 0.12.1	RMS, dBFS, ZCR, spectral centroid/rolloff, MFCC; WAV decoding
Numerics / data	NumPy 1.26.4, Pandas 2.3.3	Array math, rosters, CSV generation
Backend	FastAPI 0.143.3, Uvicorn 0.54.0, WebSockets 13.1	REST API, MJPEG stream, telemetry WebSocket
Database	SQLite3 + SQLAlchemy 2.1.5	Sessions, students, attendance, recommendations
Frontend	Streamlit 1.66.0	8-page dashboard
Config	Python-DotEnv 1.0.1	Environment configuration

2. Why These Choices

Decision	Reason
Haar cascades for detection	Fast on CPU and ships with OpenCV, with no extra model download
Custom 128-d appearance embedding for DeepSort	Avoids a heavy PyTorch re-ID network; keeps tracking light and offline
MediaPipe + solvePnP	Gives 3D head pose and iris gaze from a single RGB camera, no depth sensor needed
Facenet via DeepFace	Simple local face embeddings for consent-based recognition
FastAPI separate from Streamlit	Streamlit reruns scripts on interaction; running 30 fps vision inside it would cause frame drops and freezes
SQLite	Zero-setup embedded database, ideal for an offline single-machine app
Rule-based recommendations	Deterministic, explainable, and runs offline with no language model

3. Supported Platforms

macOS (Apple Silicon and Intel), Linux (x86_64), and Windows. Development and verification were performed on macOS with Python 3.11.12.

4. Hardware Requirements

- A standard laptop or desktop CPU; no GPU required.
- A webcam (or the built-in synthetic test video) and optionally a microphone.
- Local disk space for the SQLite file, enrolled photos, and exported CSVs.